AI BUSINESS EXPENSE & FINANCIAL DOCUMENT ASSISTANT

Finance Operations Automation Project

Main Platform: Make + Airtable

Student Group Project Guide

1. Submit Expense	2. Categorise	3. Approve & Store	4. Report
Employee enters expense and receipt	AI suggests expense category	Manager reviews and records are stored	Weekly or monthly summary is generated

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AI Business Expense & Financial Document Assistant

Detailed Student Group Project Guide | Make + Airtable

Project Summary

Students build a finance-operations workflow for small businesses. Employees submit expenses and receipt links, AI suggests categories, business rules determine whether approval is needed, records and receipts are stored, managers receive approval notifications, and a periodic expense summary is generated for management review.

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1. Project Overview

The project focuses on structured expense management rather than accounting automation. It helps students learn how to combine forms, receipts, databases, AI categorisation, approval thresholds, storage and management reporting. Human reviewers remain responsible for approving expenditure and validating financial records.

2. Project Goal

- Capture employee expense submissions.
- Create a unique expense ID.
- Store amount, vendor, date, department, purpose and receipt link.
- Use AI to suggest expense category and short description.
- Apply approval thresholds using business rules.
- Notify managers about expenses requiring approval.
- Store receipts in an organised folder or link structure.
- Track Approved, Rejected, Paid and Needs Clarification statuses.
- Generate weekly or monthly expense summaries.
- Flag missing receipts or unusual values for manual review.

3. Business Problem

Small businesses often manage expenses through chats, email, paper receipts and disconnected spreadsheets. This can result in missing evidence, unclear approval status, duplicate submissions and weak management visibility. The project creates a central expense workflow without pretending to replace accounting controls or professional financial review.

4. Project Scenario

An employee submits a NGN 185,000 expense for Facebook and Instagram advertising with a receipt. AI suggests Marketing - Paid Advertising. Because the amount is above the company's manager-approval threshold, the workflow sends an approval request, stores the receipt link and updates the record after the manager's decision.

5. Tools Required

Tool	Purpose
Make	Main workflow automation platform.
Airtable	Expense database, approval status and management views.
Tally / Google Forms	Expense submission form.
OpenAI or another AI model	Suggests expense category and summary.
Google Drive	Stores receipts and supporting files.
Gmail / Outlook	Approval notifications and employee updates.
Google Docs / Sheets (optional)	Periodic summary report.
Slack / Teams (optional)	Manager approval alerts.

6. Main Project Features

- Expense submission.
- Receipt capture / storage.
- AI expense categorisation.
- Department tagging.
- Approval threshold rules.
- Missing receipt flag.
- Duplicate-check indicators.
- Manager approval notifications.
- Expense status tracking.
- Weekly or monthly management summary.

7. Workflow Architecture

Layer	What Happens
Input	Employee submits expense and receipt information.
Storage	Expense record is created with unique ID.
AI	Expense purpose is categorised and summarised.
Validation	Rules check amount, receipt and required fields.
Approval	Expenses above threshold or flagged cases go to manager.
Storage	Receipt link and final record are kept together.
Status	Approved / Rejected / Paid / Needs Clarification is tracked.
Reporting	Periodic summary groups expenses by category and department.
Management review	Humans investigate exceptions or unusual patterns.

8. Data Capture Structure

Field	Type	Purpose
Employee Name	Short text	Person submitting expense.
Employee Email	Email	Status communication.
Department	Dropdown	Cost centre / team.
Expense Date	Date	Transaction date.
Vendor	Short text	Supplier or merchant.
Amount	Currency / number	Expense amount.
Purpose	Long text	Business reason for the expense.
Receipt	File / URL	Supporting document.
Payment Method	Dropdown	Cash, card, transfer, personal funds, etc.
Project / Client	Short text (optional)	Links expense to project.
Additional Notes	Long text	Clarification or context.

9. Database / Tracking Structure

Column	Purpose
Expense ID	Unique reference.
Employee	Submitter.
Department	Team / cost centre.
Date	Expense date.
Vendor	Supplier.
Amount	Submitted amount.
AI Category	Suggested category.
AI Summary	Short description.
Receipt Link	Stored receipt location.
Receipt Missing	Yes / No.
Approval Required	Yes / No.
Approver	Assigned manager.
Status	New / Awaiting Approval / Approved / Rejected / Paid / Needs Clarification.
Submitted At	Submission timestamp.
Approved At	Approval timestamp.
Notes	Human review comments.

10. AI Prompt Template

AI should suggest a category and summary only. It should not decide whether a transaction is tax-deductible, fraudulent, compliant or approved. Approval thresholds and finance policies must be explicit business rules.

```
You are an expense-categorisation assistant for a business.
Use the submitted expense information to suggest a category and concise summary.

Department: {{Department}}
Vendor: {{Vendor}}
Amount: {{Amount}}
Purpose: {{Purpose}}
Payment Method: {{Payment Method}}
Project / Client: {{Project Client}}

Choose one category:
Marketing
Travel
Meals
Software
Office Supplies
Training
Professional Services
Logistics
Utilities
Equipment
Other

Return:
Category:
Subcategory:
Short Summary:
Possible Duplicate Risk: Low / Review
Missing Information:
Reviewer Note:

Do not approve or reject the expense. Do not give tax, legal or audit conclusions. If the purpose is unclear, say that human clarification is required.
```

11. Automation Build

Scenario 1 - Expense Intake & Categorisation

Step	Module / Action	What It Does
1	Form / Webhook	Receive new expense submission.
2	Airtable	Create expense record and ID.
3	Google Drive	Store or link receipt file.
4	OpenAI	Suggest category, summary and review notes.
5	Make Tools / Filter	Check amount threshold and receipt presence.
6	Airtable	Update category, flags and approval requirement.
7	Router	Send to manager approval or auto-record path based on policy.
8	Email / Slack	Notify approver or employee.

Scenario 2 - Approval & Status

Step	Module / Action	What It Does
1	Airtable Watch	Detect approval decision.
2	Filter / Router	Process Approved, Rejected or Needs Clarification.
3	Airtable	Update status, reviewer and timestamp.
4	Email	Notify employee of result.
5	Optional Finance Queue	Notify finance when Approved and ready for payment.

Scenario 3 - Monthly Management Summary

Step	Module / Action	What It Does
1	Schedule	Run at end of week or month.
2	Airtable Search	Retrieve approved expenses in period.
3	Aggregator	Group by category and department.
4	OpenAI (optional)	Summarise trends using aggregated data.
5	Email / Google Docs	Send management summary.

12. Business Rules and Routing

- Use explicit amount thresholds for approvals.
- A missing receipt should flag the expense for review unless policy allows an exception.
- AI categorisation is a suggestion and can be corrected by a human reviewer.
- Do not automatically label an expense fraudulent.
- Duplicate checks should be treated as review indicators, not proof.
- Preserve original submitted values and human decision history.
- Only approved expenses should enter the payment-ready queue.

13. Sample Testing Data

Employee	Department	Vendor	Amount	Purpose	Expected Category	A
Ada	Marketing	Meta	NGN 185,000	Facebook and Instagram ads	Marketing	Y
Tunde	Operations	Bolt	NGN 18,500	Client delivery visit	Travel	N
Mariam	Product	Figma	NGN 95,000	Team design software renewal	Software	Y
Chika	HR	Coursera	NGN 60,000	Staff learning subscription	Training	N
David	Admin	OfficeMart	NGN 240,000	Printer and office equipment	Equipment	Y

14. Group Roles

Role	Responsibility
Project Lead	Defines expense process and coordinates team.
Form Designer	Builds submission experience and receipt fields.
Airtable Designer	Creates tables, views, statuses and approvals.
Prompt Engineer	Builds expense categorisation prompt.
Make Builder	Builds intake, routing, approval and reporting scenarios.
QA / Finance Reviewer	Checks thresholds, receipts and audit trail.
Presenter	Demonstrates the full expense lifecycle.

15. Expected Final Output

- Expense submission form.
- Airtable expense tracker.
- Receipt storage or link workflow.
- AI category suggestion.
- Approval threshold logic.
- Manager notification.
- Approved / rejected / clarification status path.
- Employee outcome email.
- Periodic management expense summary.
- At least five tested expense records.

16. Testing Checklist

Test Area	What to Confirm
Form	Required expense information is stored correctly.
Receipt	Missing receipt is flagged according to rule.
Category	AI produces reasonable categories but can be overridden.
Threshold	Amount above configured limit requires approval.
Approval	Rejected expense does not enter payment-ready status.
Status	All key stages have timestamps.
Duplicate risk	Similar submissions are flagged for review, not auto-rejected.
Reporting	Summary totals match the database records.
Failure handling	Invalid amount or missing employee email is identified.

17. Mini Project Assignment

Build an AI Business Expense & Financial Document Assistant using Make and Airtable. The system should capture employee expenses, store receipt links, suggest categories using AI, apply approval rules, track decisions and generate a periodic management summary.

Assignment Requirements

- Create an expense form with at least nine useful fields.
- Create an Airtable base with IDs, statuses and timestamps.
- Store or link receipts.
- Use AI to suggest category and summary.
- Create at least one amount-based approval threshold.
- Create a missing-receipt review rule.
- Build manager approval notifications.
- Update employee after approval / rejection.
- Build a weekly or monthly expense summary.
- Test at least five expenses across multiple departments.

Submission Requirements

- Screenshot of expense form.
- Screenshot of Airtable base.
- Screenshots of Make scenarios.
- Copy of categorisation prompt.
- Example approval request.
- Example employee outcome email.
- Example monthly summary.
- Testing table with expected outcomes.
- Short explanation of why AI does not make final financial decisions.

18. Presentation Guide

- Introduce the sample business and expense problem.
- Submit a sample expense live.
- Show receipt and database record.
- Show AI category suggestion.
- Explain the approval threshold.
- Demonstrate a manager decision.
- Show employee notification.
- Demonstrate one missing-receipt or duplicate-risk case.
- Show the periodic management summary.
- Explain how the workflow could integrate with a real accounting system later.

19. Assessment Rubric

Area	Weight	What Good Work Looks Like
Business understanding	15%	Clear problem definition and sensible workflow for the chosen industry.
Automation logic	25%	Reliable triggers, mappings, filters, routers, status updates and error handling.
AI implementation	20%	Useful prompt, structured output, good grounding and no unsafe business decisions delegated to AI.
Data design	15%	Clean database fields, IDs, statuses and timestamps that support tracking.
User experience	10%	Clear forms, messages, documents and notifications.
Testing	10%	Multiple realistic test cases including edge cases and failures.
Presentation	5%	Confident demo showing the complete workflow and lessons learned.

20. Possible Future Improvements

- Accounting software integration.
- Corporate card transaction matching.
- Receipt data extraction.
- Budget-versus-actual dashboard.
- Project-level cost tracking.
- Multi-level approvals.
- Foreign-currency conversion.
- Employee reimbursement workflow.
- Policy knowledge-base checks with human review.
- Management anomaly alerts based on defined business rules.

Class Outcome

Students learn how to automate a finance-operations workflow with forms, Airtable, AI categorisation, explicit approval rules, receipt handling, notifications, status tracking and management reporting.